



Department of Education

**STEM Education
National Schools of Excellence
Engineering Syllabus
Grade 12
Term 1**

**Science
Technology
ENGINEERING
Mathematics**

Issued free to schools by the Department of Education

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SECRETARY'S MESSAGE

This Engineering Syllabus is to be used exclusively by teachers of Engineering to teach Grade 12 Students enrolled for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The Grade 11 STEM Engineering Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Engineering before undertaking the advanced concepts introduced in this syllabus.

This syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principal object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and Skills and apply them to solve every day problems whilst appreciating Engineering as a fundamental field of science.

This STEM Engineering Syllabus for Grade 12 constitutes six out of the ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Engineering.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Engineering is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Engineering to human lives, including good health and environment.

In order to ensure that teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Engineering to be used for the Grade 12 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Engineering is the creative application of scientific principles to design or develop structures, machines, apparatus, or manufacturing processes. It stems from the application of applied mathematics and applied science.

Since Engineering is a practical science, engineering design process, scientific methods and applications becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Engineering.

The Grade 12 Engineering Syllabus entails six (6) of ten (1) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Engineering is a highly specialized subject students undertaking Engineering are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Engineering through the process of imparting basic Engineering knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1: Agricultural Biotechnology	1. Farming System in Russia – Krasnodar Krai 2. Agricultural Machinery for Farming 3. Agricultural Engineering – Preparing Land for Agriculture 4. Designing Markham for farming – Agricultural Engineering
HEALTH AND MEDICINE	12.1: Biomedical Engineering	1. Introduction to biomedical engineering 2. Medical Instrument 3. Applications of Biomedical Engineering 4. Case Studies of Biomedical Engineering Applications
ENERGY	12.1: Energy Generation	1. Solar Energy – Solar Power (Photovoltaic System) 2. Wind Energy 3. Hydro-Electricity Energy (Hydropower) 4. Geothermal Energy
BATTERY TECHNOLOGY	12.1: Battery Performance	1. Types of Battery 2. Battery Performance 3. Battery Life and Maintenance 4. Battery Load Testing
ARTIFICIAL INTELLIGENCE	12.1: Artificial Intelligence	1. Introduction to Artificial Intelligence 2. Fundamentals of Artificial Intelligence 3. Natural Language Processing 4. Future of Artificial Intelligence
ROBOTICS	12.1: Fundamentals	1. Robotics – The Perfect Human Beings? 2. Robotics – Hardware 3. Robotics – Software

	of Robotics Engineering	4. Constructing a robot
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RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In the STEM Grade 11 Engineering Syllabus, students were introduced to the basic concepts and knowledge of Engineering in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Engineering as they are applied to six Thema.

The introduction of the STEM Curriculum will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development, new nomenclature was introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g., Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g., Engineering, Physics, Engineering, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, Engineering and Engineering and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Engineering, Engineering and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM GRADE 12 ENGINEERING CONTENT OVERVIEW – GRADE 12, TERM 1

Table 1.1 Sequencing & Benchmarking STEM Engineering Content Standards

Setting STEM Engineering Content Standards for Progressive Learning & Assessment			
Thema: Biotechnology	Module Content Standards		
Module 12.1 Agricultural Biotechnology	12.1. Assessing the importance of engineering simple engineering techniques involved in farming for developing countries compared to developed countries.		
Topics	Performance Indicators Students are expected to:	Module Content Benchmark	Assessment Tasks
1. Farming System in Russia – Krasnodar Krai	12.1.1a: Understand that Krai is a leading production of Commercial agriculture in Russia. 12.1.1b: Identify and discuss More land is diverted into agriculture. 12.1.1c: Analyze and interpret Agriculture controls the industry's development in Krasnodar. 12.1.1d: Identify and discuss how engineering is applied. 12.1.1e: Interact and work together as a team to develop people skills.	Student's ability to master knowledge and skills of: - Understand and evaluate the importance of engineering in farming; - Evaluate the significance of machinery used in farming; - Analyze and appreciate the importance of mass farming; - Analyze and asses the developments of farming in Russia; and - Evaluate and comment on farming in developing countries – PNG.	Practical Assignment + Quiz + Test + Case Study
2. Agricultural Machinery for Farming	12.1.2a: Demonstrate and understand that machinery for farming us essential. 12.1.2b: Discuss the importance of machinery in farming. 12.1.2c: Articulate the engineering techniques involved.		
3. Agricultural Engineering – Preparing the Land for Agriculture	12.1.3a: Critically comment and assess on selection of land for agriculture. 12.1.3b: Discuss on the procedures involved in		

	preparing land for agriculture. 12.1.3c: Apply strategies and tools for preparing land.		
4. Designing Markham for Farming – Agricultural Engineering	12.1.4a: Describe and discuss on designing mass farming of biotech crops. 12.1.4b: Analyze and comment on the related procedures and concepts involved. 12.1.4c: Critically describe the need for large scale farming in PNG. 12.1.4d: Develop an Understanding on the significance of large-scale farming of crops production.		

Thema Health and Medicine	Module Content Standards		
Module 12.1: Biomedical Engineering	12.1: Develop a fundamental knowledge on the applications of biomedical engineering in Medicine.		
Topics	Performance Indicators Students are expected to:	Module Content Benchmark	Assessment Tasks
1. Introduction to Biomedical Engineering	12.1.1a: Define biomedical engineering and its related terms. 12.1.1b: Articulate the significance of biomedical engineering. 12.1.1c: Analyze the strategies and engineering tools employed by biomedical engineers. 12.1.1d: Describe and discuss the different disciplines of biomedical engineering. 12.1.1e: Outline the medical significance of the disciplines of biomedical engineering.	Students' ability to master knowledge and skills of: - Define and discuss what biomedical engineering and its concepts; - Distinguish the different types of Biomedical engineering in medical applications; and	Practical Assignment + Quiz + Test + Case Study
2. Medical Instrument	12.1.2a: Describe and discuss the purpose of medical equipment.		

	12.1.2b: Articulate and explain the steps involved in the instrumentation system. 12.1.2c: Outline the classifications of biomedical instruments. 12.1.2d: Assess the processes involved in the design process of medical instruments.	Evaluating and assessing the different application of biomedical engineering.	
3. Applications of Biomedical Engineering	12.1.3a: Assess the process involved in the applications of biomedical engineering. 12.1.3b: Assess the engineering tools employed in the processes of the applications of biomedical engineering.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Module Content Standards	12.1	12.1
Biotechnology / Module 12.1: Agricultural Biotechnology	12.1. Assessing the importance of engineering simple engineering techniques involved in farming for developing countries compared to developed countries.	✓	
Health and Medicine / Module 12.1: Biomedical Engineering	12.1: Develop a fundamental knowledge on the applications of biomedical engineering in Medicine.		✓

THEMA

BIOTECHNOLOGY

Introduction

Biotechnology combines biology with technology to engineer, modify or upgrade the parts of a biological system for industrial and human welfare. It combines various technologies derived from biology, mathematics, physics, chemistry and engineering to modify the biological systems. Its application ranges from agriculture to industry, medicine, nutrition, environmental conservation, cell biology, and so on. Biotechnology is now one of the fastest growing fields in science.

In Engineering, there are many sub-disciplines that employ Biotechnology in their profession, the most dominant of which is Genetic Engineering. These Engineers typically study DNA and manipulate them for agricultural and medical purposes. Some examples include developing genetically modified (GM) crops and animals for healthy and increased food production and the recent development of the AstraZeneca Vaccine for the COVID-19 Outbreak. Environmental and Agricultural Engineers are also integrating biological systems. By employing microorganisms into contaminated water and soil, the land will eventually be remediated for land use such as farming or for construction purposes. In essence, similar to Biomimicry, Biotechnology is not restricted to any particular discipline and has practically many applications of which some will be discussed in this Thema.

MODULE 12.1: AGRICULTURAL BIOTECHNOLOGY

5 weeks

The use of biotechnology in agriculture is focused on stable development of agricultural production, solving of food security and safety issues, obtainment of high-quality and ecologically clean food products, agricultural production waste recycling, soil fertility recovery.

PNG's agricultural growth and biotechnology in PNG Biotechnology has the ability to solve problems in PNG's agriculture production and processing in all areas. Plant breeding is used to increase and stabilize yields, improve tolerance to pests, diseases, and abiotic stresses such as drought and cold, and improve food nutritional content.

In this module, it will give a few backgrounds on the farming system in in Russia, the machinery needed for farming and how engineering contributes to into farming/ engineering agriculture in PNG and an example on how we can cultivate our land mass

Linking Module

- STEM-proper Grade 11/ Engineering Phylum / Thema Biotechnology / Module 11.1: Introduction to Biotechnology
- STEM-proper Grade 11/ Biology Phylum / Thema Biotechnology / Module 11.1: Concepts and Applications of Biotechnology /Topic 2: Industrial Biotechnology

Module Content Standard

12.1. Assessing the importance of engineering simple engineering techniques involved in farming for developing countries compared to developed countries.

Module Content Benchmark

Student's ability to master knowledge and skills of:

- Understand and evaluate the importance of engineering in farming;
- Evaluate the significance of machinery used in farming;
- Analyze and appreciate the importance of mass farming;
- Analyze and asses the developments of farming in Russia; and
- Evaluate and comment on farming in developing countries – PNG.

Topic 1: Farming system in Russia - Krasnodar Krai

The agro-industrial complex, which is one of Russia's largest, is the foundation of the Krai's economic potential. A krai is a sort of Russian federal subject. There are 85 federal subjects in the nation, nine of which are krais. The distinction between a political unit with the term "krai" or "oblast" is entirely conventional, comparable to the commonwealths in the United States; both are component body's equivalent in legal standing in Russia with representation in the Federation Council. Autonomous oblasts may be subordinated to republics or krais during the Soviet era, but not to oblasts. Both words have a fairly similar general meaning ("region" or "area") outside of political vocabulary, and they are frequently interchanged.

The food sector is the Krai's main industry, accounting for 42.8 % industrial production. Textile industry is also featured among the light industries (kamgar-wool Corporation and carpet manufacturing in Krasnodar). The Krai is also a leader in the production of commercial agricultural. Agriculture accounts for 16 percent of overall GDP (7.8% in Russia), and it is the country's most significant agricultural area, ranking first in Russia and accounting for 7% of total gross agriculture production. Winter wheat, rice, maize, sunflower, sugarcane, and castor oil plant are the most significant crops. The overall wheat production (10% of Russia's gross crop) and sugarcane yield (17.3%) are the highest in the country. In addition, the Krai is a leader in the production of sunflowers and wine (15 percent and 37 percent respectively). The North Caucasus area is Russia's largest rice producer, with Kuban as one of its major production hubs. Aside from tea, which is cultivated in the south, plant oil production, which accounts for one-third of Russia's total output, as well as sugar (one-third of total output) and meat production, is also significant. The Kuban delta has a thriving fishing industry.

Performance Indicators

Students are expected to:

12.1.1a: Understand that Krai is a leading production of Commercial agriculture in Russia.

12.1.1b: Identify and discuss More land is diverted into agriculture.

12.1.1c: Analyze and interpret Agriculture controls the industry's development in Krasnodar.

12.1.1d: Identify and discuss how engineering is applied.

12.1.1e: Interact and work together as a team to develop people skills.

Content Expansion

Body of Knowledge for their Understanding

- Krasnodar Krai agro-industrial region
- The Structure of agriculture
- Grain production in Krasnodar Krai
- Viticulture
- Horticulture
- Gardening
- Farming
- Program is the effective development of the agro-industrial complex of Krasnodar region

Set of Skills for their Proficiency Levels

- Recognize the significance of farming and the engineering techniques involved.

Trait of Attitudes and Values to govern their lives

- Recognize the significance of farming and the engineering techniques involved.

Topic 2: Agricultural Machinery for Farming

The mechanical structures and devices that are utilized in farming and other forms of agriculture are referred to as agricultural equipment. Many varieties of farming equipment as well as tractors, hand tools and power tools that tow or control, are all examples of farming equipment. Both organic and nonorganic farming employ a wide range of equipment. Agricultural technology has become an important element of how the world gets fed, especially with the introduction of mechanized agriculture. This topic will discuss the farm machinery types, its uses and their importance's. ¹

Performance Indicators

Students are expected to:

12.1.2a: Demonstrate and understand that machinery for farming is essential.

12.1.2b: Discuss the importance of machinery in farming.

12.1.2c: Articulate the engineering techniques involved.

Content Expansion

Body of Knowledge for their Understanding

- Introduction
- Tractor
- Sprayer
- Shredders and cutters
- Machinery used for soil cultivation
 - Cultivator
 - Cultipacker
 - Plough

Set of Skills for their Proficiency Levels

- Recognize and understand the operations of machinery used for farming.

Trait of Attitudes and Values to govern their lives

- Appreciate the engineering skills and techniques involved in the farming operations through the use of machinery.

¹ (Encyclopedia, 2021)

Topic 3: Agricultural Engineering - Preparing Land for Agriculture

The art and science of cultivating the land, growing crops, and raising livestock is known as agriculture. It entails preparing plant and animal goods for human consumption as well as their delivery to markets. Engineering agriculture deals with farm machinery design, farm structure location and planning, farm drainage, soil management and erosion control, water supply and irrigation, rural electricity, and farm product processing. This topic will discuss on the how best preparation of land should be done in order to have agricultural farming.

Performance Indicators

Students are expected to:

12.1.3a: Critically comment and assess on selection of land for agriculture.

12.1.3b: Discuss on the procedures involved in preparing land for agriculture.

12.1.3c: Apply strategies and tools for preparing land.

Content Expansion

Body of Knowledge for their Understanding

- Land Preparation
- Methods of land clearing
- Importance of land demarcation
- Determine a land size.
- Sketch layout of farm land.
- Line and peg farm land
- Ploughing and harrowing
- Prepare ridges
- Importance of soil amendments
- Laying plastic mulch

Set of Skills for their Proficiency Levels

- Understand, recognize and assess the process involved in preparing the land for Agriculture.

Trait of Attitudes and Values to govern their lives

- Appreciate the procedures and steps involved in preparing a land.

Topic 4: Designing Markham for Farming – Agricultural Engineering

Papua New Guineans settled on the land and became farmers, cultivating the land and utilizing the abundant biodiversity for food and feed over 9,000 years ago while the Europeans were still hunters and gatherers. The people in the Kuk Valley in Western Highlands Province started the global Agricultural Civilization where drainage, planting, digging and staking of plants have been used to cultivate taro, banana and yam between 7,000 – 10,000 years ago. These agricultural practices have propagated throughout different societies that settled in PNG dating back to 9,000 years. As a result, PNG has the world's largest agricultural biodiversity and therefore the world's largest agricultural gene pool.

As a result, Papua New Guinea has developed into a very rich agricultural biodiversity hotspot in the world, with farmers growing over 200 species of plants for food and feed. At present, of the country's seven million people, 70% depend on agriculture.

This topic will give students ideas on how they can cultivate land for farming using the Markham plain as an example. Throughout the topic student should be able to plan and design on how they can relocate villagers, get their possible power supplies, do irrigations systems and produce large scale productions on cash crops.

Performance Indicators

Students are expected to:

12.1.4a: Describe and discuss on designing mass farming of biotech crops.

12.1.4b: Analyze and comment on the related procedures and concepts involved.

12.1.4c: Critically describe the need for large scale farming in PNG.

12.1.4d: Develop an Understanding on the significance of large-scale farming of crops production.

Content Expansion

Body of Knowledge for their Understanding

- Farm layout
- Plan for farming large scale production
- New Road Link
- Redirecting waterway
- Dams
- Land for Cultivation
- Relocation of Villagers
- Plantation farming system Farming system
- Land for Agriculture
- Machinery for farming
- Irrigation
- Power Supply

Set of Skills for their Proficiency Levels

- Recognize the significance of farming and the engineering techniques involved.

Trait of Attitudes and Values to govern their lives

- Appreciate agricultural biotechnology and its applications on farming.

THEMA HEALTH AND MEDICINE

Introduction

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

Health and Safety will always be mentioned and emphasized in the field of Engineering. Regardless of the type of work and the level of risks involved that may or may not cause injury or bodily harm, the intensity of enforcing Health and Safety will not lessen. It is an ethical conduct and cannot be stressed enough. A daily dosage of danger and safety related issues will always be encountered in the life of an engineer, particularly those out in the field. These dangerous encounters can include nuclear plant explosions, building collapses during construction and even underwater landslides for marine engineers all. This is why Health and safety will always be a habitual routine and practice that all engineers must retain.

Engineering also has contributed greatly in the field of health and medicine, particularly in artificial creations of limbs and organs (i.e., prosthetics) as well as the development of medical equipment such as X-rays and Magnetic Resonance Imaging (MRI) that have improved the efficiency of the delivery of Medical Services. The collaborative efforts of engineers and scientists continually raise the effectiveness of medical service delivery and boost the standards of human welfare.

MODULE 12.1: BIOMEDICAL ENGINEERING

4-5 weeks

In this module, the application of engineering principles and problem-solving techniques in the field of medicine will be discussed. It also focuses on improving the health care for human beings at all level. An introduction to the module followed by the disciplines and application of biomedical engineering will be discussed in the content.

Linking Module

- STEM Bridging / Engineering / Module 11.1: Introduction to Engineering
- STEM Bridging / Engineering / Module 11.2: Introduction to Engineering Design Process
- STEM Bridging / Engineering / Module 11.1: Engineering in Health and Medicine

Module Content Standard

12.1: Develop a fundamental knowledge on the applications of biomedical engineering in Medicine.

Module Content Benchmark

Students' ability to master knowledge and skills of:

- Define and discuss what biomedical engineering and its concepts;
- Distinguish the different types of Biomedical engineering in medical applications; and
- Evaluating and assessing the different application of biomedical engineering.

Topic 1: Introduction to Biomedical Engineering

Medical engineering also known as biomedical engineering utilizes engineering principles and problem-solving techniques to medicine. This topic introduces to students on biomedical engineering, the significance of biomedical engineering and the role of being a biomedical engineer.

Performance Indicators

Students are expected to:

12.1.1a: Define biomedical engineering and its related terms.

12.1.1b: Articulate the significance of biomedical engineering.

12.1.1c: Analyze the strategies and engineering tools employed by biomedical engineers.

12.1.1d: Describe and discuss the different disciplines of biomedical engineering.

12.1.1e: Outline the medical significance of the disciplines of biomedical engineering.

Content Expansion

Body of Knowledge for their Understanding

- What is Biomedical engineering?
- Role of a biomedical engineer?
- Importance of biomedical engineering
- Disciplines of Biomedical Engineering

Set of Skills for their Proficiency Levels

- Understand and be able to explain the concept of biomedical engineering to medicine.
- Develop an Understanding of the disciplines of biomedical engineering.

Trait of Attitudes and Values to govern their lives

- Identify the role of engineering principles and problem-solving techniques to medicine.

Topic 2: Medical Instrument

This topic will outline the general concepts of medical devices that are applicable to all medical instrumentation systems. A medical device/instrument is a no biochemical object used in human patients for diagnostic, therapeutic, supporting, replacing, prevention, or any other physical effect-producing application. In Papua New Guinea medical equipment is both a capital investment and an expensive asset.

Performance Indicators

Students are expected to:

12.1.2a: Describe and discuss the purpose of medical equipment.

12.1.2b: Articulate and explain the steps involved in the instrumentation system.

12.1.2c: Outline the classifications of biomedical instruments.

12.1.2d: Asses the processes involved in the design process of medical instruments.

Content Expansion

Body of Knowledge for their Understanding

- Definition of Medical Instrument or Device
- Generalized Medical instrument system
- Medical instrumentation Modes
- Medical measurement constraints
- Classifications of Biomedical instruments
- Interfering and modifying inputs
- Design Criteria and Process

Set of Skills for their Proficiency Levels

- Analyze and distinguish the components of medical instruments.

Trait of Attitudes and Values to govern their lives

- Identifying the role of different components of medical instrument systems and its relation to engineering.

Topic 3: Applications of Biomedical Engineering

This topic discusses on the different applications involved in the biomedical engineering. It covers some of the healthcare purposes in medicine and biology through therapeutic and diagnostic applications. Some of the prominent biomedical engineering applications include the development of:

- biocompatible prostheses;
- various diagnostic and therapeutic medical devices ranging from clinical equipment to micro-implants;
- common imaging equipment such as MRIs and EEGs;
- regenerative tissue growth;
- pharmaceutical drugs; and
- therapeutic biologicals.

Performance Indicators

Students are expected to:

12.1.3a: Assess the process involved in the applications of biomedical engineering.

12.1.3b: Asses the engineering tools employed in the processes of the applications of biomedical engineering.

Content Expansion

Body of Knowledge for their Understanding

- Application-specific strategies for the design and selection of biomaterials
- BIOMATERIALS: TYPES, PROPERTIES, AND THEIR APPLICATIONS
- Magnetic Resonance Imaging
- Electrocardiography
- The Ambulatory ECG
- Patient Monitoring
- High-Resolution ECG

Set of Skills for their Proficiency Levels

- Recognize the engineering tools employed.
- Identify and understand the significance of integrating engineering knowledge in medical applications.

Trait of Attitudes and Values to govern their lives

- Compare and evaluate the important disciples of engineering employed in the applications.

Topic 4: Case Studies of Biomedical Engineering Applications

These are case studies of how engineering designs were used to solve medical problems through innovations either from natural inspirations or out of curiosity and creativeness. The advancing technology has created a platform for new innovations in how engineering designs would be utilized effectively to solve medical problems. It is a challenge for the upcoming generation of biomedical engineers to fully maximize and utilize technology to engineer new innovative designs to solve pending medical issues.

Performance Indicators

Students are expected to:

12.1.4a: Clearly define and explain foot drop and the engineering solutions to it.

12.1.4b: Understand and analyze the process involving engineering and medicine for the solution

12.1.4c: Discuss and assess different engineering solutions to medical problems that are impending

Content Expansion

Body of Knowledge for their Understanding

- Engineering Solutions to Foot Drop
- Anatomy and Physiology of Foot Drop
- Gait Cycle
- Dorsiflexion Importance
- Treatments
- Engineering Solutions
- Important things to consider in designing an AFO
- Biomechanics Of Foot Drop in Relation to Designing an AFO
- Effectiveness Of Ankle Foot Orthoses
- The Electrical Approach - Functional Electrical Stimulation

Set of Skills for their Proficiency Levels

- Develop a creative and analytical skill to solve medical problems

Trait of Attitudes and Values to govern their lives

- Recognize the importance of applying engineering to solve medical problems.

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